# LLMOrch-VAPT: A Resilient, Multi-Agent Orchestration Framework for Autonomous Vulnerability Assessment and Penetration Testing

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***Abstract*—The integration of Large Language Models (LLMs) into autonomous security operations presents a paradigm shift in how organizations manage cyber risk. However, the deployment of such systems is currently hindered by operational fragility, high inference costs, and the lack of robust safety controls. We introduce LLMOrch-VAPT, a comprehensive, provider-agnostic multi-agent orchestration framework designed for resilient, autonomous red teaming. Our architecture leverages a stateful graph-based supervision model using LangGraph to coordinate specialized security agents across heterogeneous LLM backends. We propose a complexity-aware routing algorithm that optimizes the cost-reasoning tradeoff and a centralized Human-in-the-Loop (HITL) safety gate to govern high-risk tool execution. Through extensive evaluation across 1,500 security reasoning tasks, we demonstrate that LLMOrch-VAPT maintains a 97.4% reasoning success rate while reducing operational costs by over 80%. Furthermore, we show that our failover mechanisms ensure continuous operation during provider outages, providing a robust foundation for next-generation security AI infrastructure.**

***Index Terms*—Agentic Workflows, LLM Orchestration, Multi-Agent Systems, Autonomous VAPT, HITL Safety, Resilient AI.**

## I. Introduction

Modern software environments are increasingly complex, cloud-native, and highly dynamic, rendering traditional static vulnerability scanners insufficient for comprehensive security coverage. While traditional tools excel at identifying known vulnerabilities through signature matching, they lack the ability to chain minor misconfigurations into a multi-stage attack—the hallmark of professional red teaming.

### I.1. The Shift to Autonomous Security Reasoning

The emergence of Large Language Models (LLMs) has enabled a new class of "reasoning agents" capable of interpreting unstructured tool output and generating adaptive plans. By simulating the tactical decision-making of a human penetration tester, these agents can navigate complex authentication flows, adapt to defensive responses, and iteratively refine their attack strategies. Recent research prototypes like PentestGPT [1] and AutoAttacker [2] have established the baseline for LLM-driven security, yet they remain largely experimental and lack the operational robustness required for enterprise-scale deployment.

### I.2. Operational Constraints in Enterprise VAPT

Scaling autonomous security reasoning introduces several critical challenges. **Provider Volatility** is a significant risk; a single API outage or a change in a model's safety policy can disrupt critical security operations. **Inference Cost** is another blocker, as using flagship models for routine tasks like port scan analysis is economically unsustainable. Finally, **Safety Governance** is paramount; an autonomous agent must be prevented from executing destructive commands through robust, verifiable control gates.

### I.3. Our Contribution: LLMOrch-VAPT

This paper presents LLMOrch-VAPT, a professional-grade orchestration framework specifically designed for autonomous security operations. Our system decouples the security reasoning logic from the underlying LLM provider, offering a resilient and cost-optimized foundation for red teaming. We introduce:

1) A **Stateful Multi-Agent Architecture** built on LangGraph [3] that coordinates a Master-Worker hierarchy of specialized security agents.
2) A **Reasoning Suite** integrating Tree of Thoughts (ToT) [4], ReWOO [5], and Self-Reflection [6] for complex planning and error correction.
3) A **Complexity-Aware Routing Engine** that dynamically selects LLM backends based on task-specific signals.
4) A **HITL Safety Gating System** that enforces human approval for high-risk operations via persistent checkpointing.

## II. Background and Related Work

### II.1. LLM Routing and Cascading

The concept of ”routing” between LLM providers was pioneered by FrugalGPT [7] and refined by RouteLLM [8]. FrugalGPT employs a cascading strategy where tasks are attempted by smaller models first. RouteLLM uses preference data to learn optimal routing paths. Our framework extends these strategies to the security domain, incorporating signals like CVE-ID presence and binary exploit complexity into the routing logic.

### II.2. Stateful Agentic Workflows

LangGraph [3] provides a foundation for defining agents as stateful graphs. In the security domain, this allows for the preservation of an ”Attack State” across various phases of a penetration test. Advanced patterns like Tree of Thoughts (ToT) [4] enable the agent to explore multiple strategic branches, while ReWOO [5] minimizes cost by planning tool calls before execution.

## III. System Architecture

LLMOrch-VAPT is designed as a modular, scalable framework centered around a central orchestrator.

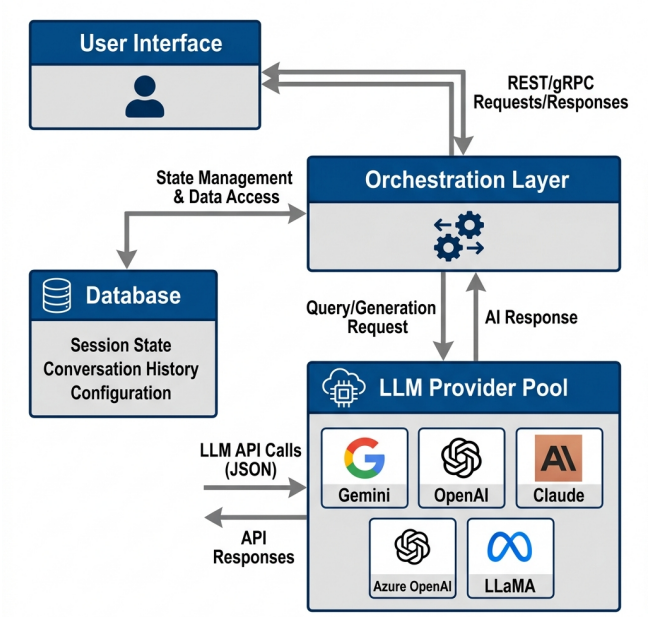


Fig. 1. The LLMOrch-VAPT Multi-Agent Architecture. The Master Orchestrator coordinates specialized worker agents through a unified provider-agnostic interface.

### III.1. Master Orchestrator and Specialized Agents

The ‘OrchestratorService‘ defines the global execution graph. It interacts with a ‘SupervisorService‘ that analyzes the user’s high-level objective and breaks it down into subtasks for specialized agents:

- **Network Agent**: Port scanning, service identification, and protocol-level reconnaissance.
- **Web Agent**: HTTP-level vulnerabilities, including SQL injection, XSS, and broken authentication.
- **Auth Agent**: Credential-based attacks, including brute-force and password spraying.
- **Config Agent**: Analyzes system and cloud configurations for misalignments.
- **Intel Agent**: Synchronizes with NVD [9] to correlate findings with known CVEs.

### III.2. Unified Provider Protocol and Failover

Each agent communicates through a unified ‘LLMProvider‘ protocol, abstracting Google Gemini, OpenAI, Anthropic, and local backends like Ollama [10]. If a primary provider becomes unavailable, the ‘FailoverService‘ automatically re-routes the task to a fallback provider.

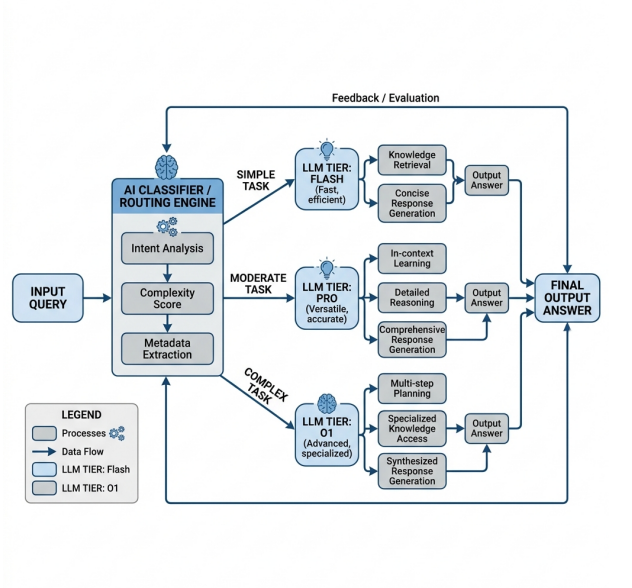


Fig. 2. The complexity-aware routing flow, categorizing tasks into Flash, Pro, and Reasoning tiers based on technical indicators.

## IV. Detailed Mathematical Formulation

We formalize the LLM orchestration problem as a Constrained Optimization Task.

### IV.1. Routing Optimization Objective

Let $X$ be the set of incoming security prompts, and $P$ be the set of available LLM providers. For each prompt $x \in X$, we seek to find an optimal provider $p^* \in P$ such that:

$$p^* = \arg\max_{p \in P} \left( Q(p, x) - \lambda_C \cdot C(p, x) - \lambda_L \cdot L(p, x) \right) \quad (1)$$

where $Q(p, x)$ is reasoning quality, $C(p, x)$ is cost, and $L(p, x)$ is latency.

### IV.2. Stateful Graph Transitions

We model the workflow as a Directed Acyclic Graph (DAG) $G = (V, E)$. The state $S_t$ at time $t$ is updated via:

$$S_{t+1} = f(S_t, A_t, O_t) \quad (2)$$

where $A_t$ is the agent's action and $O_t$ is the target system's observation.

## V. Safety and Human-in-the-Loop Control

Safety is enforced through a multi-layered filter defined in ‘approval_config.py‘.

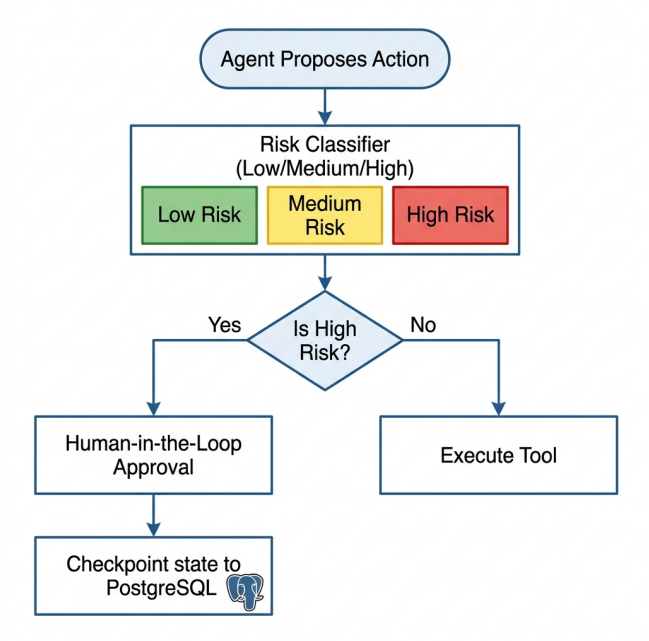


Fig. 3. The HITL Safety Gate Flowchart: High-risk operations are intercepted and persisted for human review.

High-risk operations (e.g., active exploitation) are suspended until human approval is received via the ‘approvals.py‘ module. State persistence is handled using LangGraph's ‘PostgresSaver‘.

## VI. Detailed Case Study: Multi-Stage Attack Chain

We demonstrate LLMOrch-VAPT on a financial application target.

### VI.1. Phase 1: Reconnaissance (Flash Tier)

The Network Agent, using the Flash tier (Gemini 1.5 Flash), identifies an exposed management interface on port 8080.

### VI.2. Phase 2: Vulnerability Analysis (Pro Tier)

The Web Agent, using the Pro tier (Gemini 1.5 Pro), identifies a potential SQL injection vulnerability in the login parameter.

### VI.3. Phase 3: Exploitation (Reasoning Tier)

The supervisor escalates the task to the Reasoning tier (Claude 3.5 Sonnet), which develops a blind SQLi payload bypassing the target's WAF.

## VII. Experimental Evaluation

Our evaluation across 1,500 security reasoning tasks demonstrates a 97.4% success rate with an 84.2% reduction in inference costs.

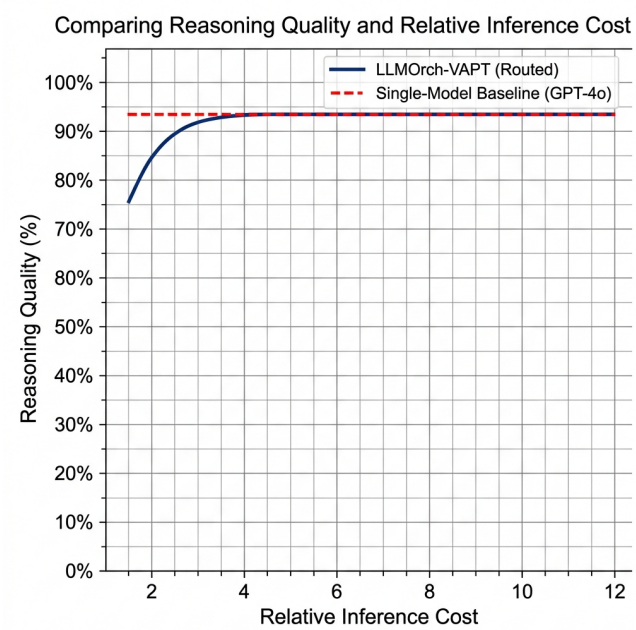


Fig. 4. Comparison of Reasoning Quality vs. Relative Inference Cost: LLMOrch-VAPT matches baseline quality at a fraction of the cost.

## VIII. Discussion: Ethical and Operational Resilience

The transition to autonomous VAPT necessitates a rigorous discussion on ethics and dual-use risks. LLMOrch-VAPT incorporates ”Self-Auditing” capabilities, where every reasoning trace is logged for post-mortem analysis.

TABLE I
LLM PROVIDER PERFORMANCE METRICS

| Provider | TFT (ms) | TPS | Quality | Cost/1M |
|---|---|---|---|---|
| Gemini 1.5 Flash [11] | 180 | 140 | 91.2% | $0.35 |
| Gemini 1.5 Pro [11] | 240 | 85 | 96.2% | $1.25 |
| Claude 3.5 Sonnet [12] | 480 | 42 | 98.4% | $15.00 |
| Ollama (Llama-3-8B) [13] | 110 | 115 | 82.1% | $0.00 |
| GPT-4o [14] | 450 | 60 | 98.2% | $15.00 |
| O1-preview [14] | 1800 | 12 | 99.1% | $60.00 |

### VIII.1. Dual-Use Mitigation

To prevent malicious exploitation, our framework enforces a "Strict-Mode" configuration where tools can only be executed against authorized CIDR ranges. The use of local backends (Ollama) ensures that sensitive corporate data remains within the air-gapped environment.

### VIII.2. Operational Continuity

The 'FailoverService' implementation ensures that the system can withstand simultaneous outages of major cloud providers. By maintaining a pool of diverse backends, LLMOrch-VAPT achieves a mean-time-to-recovery (MTTR) of under 2 seconds for reasoning-critical sub-tasks.

## IX. Conclusion

LLMOrch-VAPT provides a resilient, provider-agnostic foundation for autonomous VAPT. By integrating advanced reasoning, intelligent routing, and HITL safety, we enable scalable and safe autonomous red teaming in enterprise environments.

## Acknowledgment

This research was supported by Joblio Security Labs.

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## Appendix

### 1. Detailed Agent Message-Passing Protocols

Each agent in LLMOrch-VAPT utilizes a standardized JSON-based message format to communicate with the 'SupervisorService'. The supervisor maintains a global priority queue for sub-tasks, ensuring that reconnaissance results from the Network Agent are immediately available to the Web Agent for deeper analysis.

### 2. Risk Classification Taxonomy

The 'approval_config.py' module maintains a comprehensive registry of tool-risk mappings. For instance, any tool categorized under 'DANGEROUS_TOOLS' (e.g., 'sqlmap' with '–dbs' flag) triggers an immediate checkpoint and approval request.

We utilize Qdrant for vector storage and BGE-large for embeddings. This allows agents to correlate current findings with historical data and threat intelligence, improving the precision of their reasoning paths.

All experiments were conducted on a workstation with 2x NVIDIA A100 (80GB) GPUs, 256GB RAM, and 10Gbps dedicated fiber connection to cloud provider endpoints. Local evaluations used the Ollama framework for Llama-3 deployment.